Report 2: UML Class Diagram Report

Hotel Booking System

Course / Assignment	UML Class Diagram Modelling Assignment
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1. The UML Class Diagram

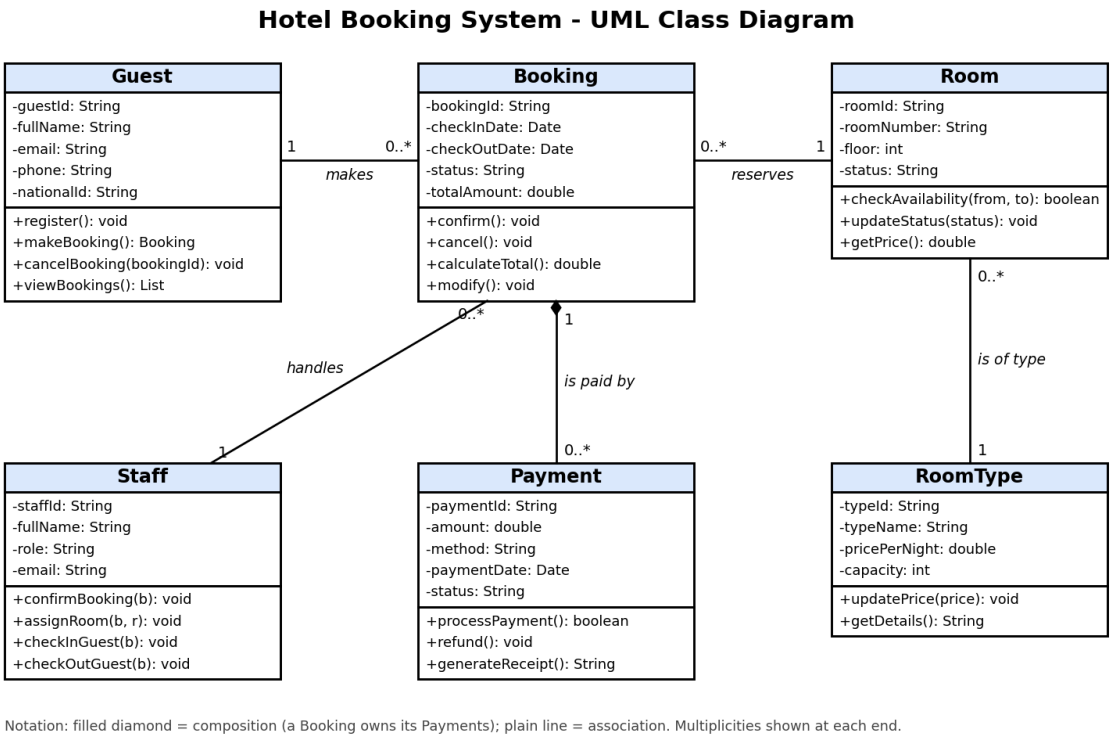


Figure 1: UML class diagram of the Hotel Booking System (created with draw.io / matplotlib; editable file in /diagrams).

2. Classes, Attributes and Methods

Guest

Attributes	Methods (operations)
-guestId: String -fullName: String -email: String -phone: String -nationalId: String	+register(): void +makeBooking(): Booking +cancelBooking(bookingId): void +viewBookings(): List

Booking

Attributes	Methods (operations)
-bookingId: String -checkInDate: Date -checkOutDate: Date -status: String -totalAmount: double	+confirm(): void +cancel(): void +calculateTotal(): double +modify(): void

Room

Attributes	Methods (operations)
-roomId: String -roomNumber: String -floor: int -status: String	+checkAvailability(from, to): boolean +updateStatus(status): void +getPrice(): double

Staff

Attributes	Methods (operations)
-staffId: String -fullName: String -role: String -email: String	+confirmBooking(b): void +assignRoom(b, r): void +checkInGuest(b): void +checkOutGuest(b): void

Payment

Attributes	Methods (operations)
-paymentId: String -amount: double -method: String -paymentDate: Date -status: String	+processPayment(): boolean +refund(): void +generateReceipt(): String

RoomType

Attributes	Methods (operations)
-typeId: String -typeName: String -pricePerNight: double -capacity: int	+updatePrice(price): void +getDetails(): String

3. Relationships and Multiplicities

Class A	Mult. A	Relationship	Mult. B	Class B	Type
Guest	1	makes	0..*	Booking	Association
Booking	0..*	reserves	1	Room	Association
Staff	1	handles	0..*	Booking	Association
Booking	1	is paid by	0..*	Payment	Composition
Room	0..*	is of type	1	RoomType	Association

4. Explanation of the Relationships

Guest - Booking (1 to 0..*, association 'makes')

One guest can make many bookings over time, or none yet if newly registered. Each booking belongs to exactly one guest.

Booking - Room (0..* to 1, association 'reserves')

Each booking reserves exactly one room, while one room can appear in many bookings (for different dates) or none.

Staff - Booking (1 to 0..*, association 'handles')

A staff member handles many bookings (confirmation, room assignment, check-in/out). Each booking is handled by one staff member.

Booking - Payment (1 to 0..*, composition 'is paid by')

A booking can have zero payments (not yet paid), one, or several (deposit and balance). Payments cannot exist without a booking, so this is a composition: if the booking record is deleted, its payments are deleted with it.

Room - RoomType (0..* to 1, association 'is of type')

Every room has exactly one room type (e.g. Single, Double, Suite), while a room type groups many rooms and sets the price per night and capacity.

5. Notation Used

• **Visibility:** '-' means private, '+' means public.

• **Multiplicity:** '1' exactly one; '0..*' zero or more.

• **Composition:** filled diamond on the 'whole' side (Booking).

• **Association:** plain line with a verb label describing the relationship.